



ImbizoHub

Technical & Systems Overview

Product, architecture, infrastructure and data handling

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imbizohub.com

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1. What ImbizoHub is

ImbizoHub is a Zimbabwean online marketplace. It brings four things people already do informally — through WhatsApp groups, noticeboards and word of mouth — into one place, with a record of who did what:

- Buying and selling goods — phones, vehicles, furniture, appliances and more
- Posting what you are looking for, so sellers come to you rather than you searching
- Hiring transport — customers post a trip, operators compete with quotes
- Book & Deliver — parcel delivery between buyer and seller (built, currently paused; see 8.2)

It runs as an Android application and as a website, from a single codebase. The commercial model is a platform fee on specific actions — unlocking a seller's contact details, accepting a transport quote, operator registration — rather than a commission on the goods themselves. Money for the goods or the trip passes directly between the two people; ImbizoHub is not in the middle of it.

Launch promotion. All platform fees are waived until 31 January 2027. The payment infrastructure is live and working; the amounts are currently set to zero.

2. Architecture at a glance

Three parts: one application, one backend, and a small number of outside services. Everything except the payment relay runs on managed infrastructure, which is a deliberate choice — it keeps the number of servers to look after at one.

Layer	Technology	Purpose
Application	Expo SDK 56, React Native 0.85, React 19	One codebase producing both the Android app and the website
Routing	Expo Router 56	45 screens; file-based routing shared across app and web
Backend	Supabase (PostgreSQL 17.6)	Database, authentication, file storage, realtime, serverless functions
Web hosting	Vercel	Landing site, web app, two serverless functions
Payments	Paynow (incl. EcoCash)	Card and mobile money, via a relay (see 5)
Notifications	Expo Push Service	Push to Android devices
Email	Namecheap Private Email	Outbound mail from the contact form

The application talks to Supabase directly for almost everything. It never holds a privileged key — the public anonymous key is all that ships, and what it can reach is controlled inside the database itself.

3. Backend — Supabase

Item	Detail
Project region	eu-west-1 (Ireland)
Database	PostgreSQL 17.6
Created	27 June 2026
Edge functions	11 deployed
Scheduled jobs	4, via pg_cron
Storage buckets	3 (see below)

3.1 Storage

Bucket	Visibility	Holds
avatars	Public	Profile photographs
listing-photos	Public	Photographs of items for sale
verification-documents	Private	National ID documents and affidavits

The third bucket is deliberately private. Identity documents are the most sensitive data the platform holds and are not reachable by URL, unlike the other two.

3.2 Scheduled jobs

Four jobs run on a schedule inside the database. Three of them enforce the published data retention policy; nothing depends on anyone remembering to run them.

Job	Schedule	What it does
delete-expired-accounts	Daily 03:00	Purges accounts 30 days after a deletion request
cleanup-expired-data	Daily 04:00	ID documents, stale push tokens, old photos, expired payment records
notify-registration-expiring	Daily 08:00	Warns operators before their registration lapses
cleanup_inactive_messages	Sun 02:00	Removes conversations inactive for two years

4. Web — Vercel and the domains

The website and the web version of the app are served by Vercel from the same repository as the mobile app. A build step exports the Expo web bundle and assembles it alongside the marketing pages.

Address	Serves
imbizohub.com	Landing site — serves directly, no redirect
www.imbizohub.com	Redirects (308) to the apex
imbizohub.com/app	The full application in the browser
imbizohub.com/contact	Contact form
imbizohub.com/data-retention	Published retention policy
imbizohub.com/seller	Shareable seller profile, server-rendered
imbizohub.app	Secondary domain; hosts password-reset and legal pages

Why the apex serves directly. Android App Links verification requires the domain-verification file to be reachable without a redirect — Google’s checker refuses to follow one. Putting `www` in front of the apex would break deep linking into the app.

4.1 Serverless functions

- `api/contact.js` — receives contact form submissions, stores them, then emails them
- `api/seller.js` — re-serves the seller preview page with the correct content type, so link previews work on WhatsApp and Facebook

4.2 Domains and email

Item	Provider
imbizohub.app — registrar	Namecheap
imbizohub.com — registrar	Namecheap
DNS	Namecheap, pointing at Vercel
Mailbox (support@imbizohub.com)	Namecheap Private Email
Legal pages	GitHub Pages

5. Payments — Paynow and the relay

Payments are handled by Paynow, which covers both card payments and EcoCash. ImbizoHub never sees a PIN, a card number or any payment credential; the user completes payment on Paynow’s own checkout page.

5.1 Why there is a VPS

This is the one piece of infrastructure that is not managed, and it exists for a specific reason. Paynow’s servers reset the connection when called from Supabase’s cloud IP ranges — confirmed by live testing and by

Paynow's own support. Requests therefore go out through a small relay running on a DigitalOcean droplet, which forwards them to Paynow and returns the response.

The relay holds no data. It signs nothing and stores nothing — it forwards a request and returns a reply. Paynow credentials stay in Supabase secrets and never reach the client.

5.2 Payment flow

- The app calls the create-payment function — it never writes a “paid” record itself
- A pending record is created with our own reference
- The request is signed and sent to Paynow through the relay
- The user is sent to Paynow's checkout page
- Paynow calls back to the paynow-webhook function when payment completes
- Only then is anything marked paid, and only server-side

The separation matters: nothing the application says can mark a payment as received. Confirmation only ever arrives from Paynow, to a server the application cannot impersonate.

6. Mobile builds and releases

Channel	Tool	Reaches
Android binary	EAS Build	Google Play — needed for native changes
Over-the-air update	EAS Update	Installed apps — JavaScript changes only
Website	Vercel, on git push	imbizohub.com
Database	Supabase migrations	Applied directly

Two consequences worth being clear about. A JavaScript fix reaches phones without a Play review, usually within minutes. Anything native — icons, permissions, deep-link configuration — requires a new build and a new Play submission.

7. Security and data handling

7.1 Access control

Two independent gates sit in front of every table. Row Level Security decides which rows a user may touch; PostgreSQL grants decide which columns. Both must permit an action for it to happen.

- Users can only read messages they sent or received
- Profile fields that must not be self-assigned — administrator status, verified status, whether a registration is paid — are not writable by users at all, regardless of what the application sends
- Contact form submissions cannot be read with the public key under any circumstances; they are written through a function that returns nothing readable

- Identity documents live in a private bucket, not a public URL

7.2 Data handling

- IP addresses are stored only as a salted one-way hash, never in the clear
- Payment credentials never reach ImbizoHub systems
- Retention periods are enforced automatically (see 3.2) and published
- A security incident log records what happened, whether POTRAZ was notified, and whether affected users were told

7.3 What is not collected

Stated plainly because many comparable apps do collect these:

- No device or GPS location — the app never requests location permission
- No analytics, crash reporting or advertising identifiers
- No third-party tracking, and no advertising of any kind

8. Current status

8.1 Live

- Website, web application and contact form
- Buying and selling, wanted posts, transport hire
- Payments, with fees waived under the launch promotion
- Android application submitted to Google Play

8.2 Built but paused

Book & Deliver is complete and switched off behind a feature flag. The reason is deliberate: there is no escrow, so a buyer and seller in different cities have no recourse if either side fails to deliver. New delivery-operator registration is closed with it. Re-enabling is a one-line change once a payment-protection flow exists.

8.3 Known gaps

Item	Note
iOS application	Not built — requires an Apple Developer account
Escrow	Not built; the reason Book & Deliver is paused
Paynow relay transport	Runs over plain HTTP on an IP address and port. Protected by a shared secret, but the traffic itself is unencrypted. Worth putting behind TLS.
Operational monitoring	No crash reporting or alerting. Good for privacy, but faults are found by report rather than by alarm.

Data residency

The database is hosted in Ireland. Relevant to any cross-border transfer assessment under the Cyber and Data Protection Act.

9. Suppliers

Everyone who holds or processes ImbizoHub data, in one place. This list is also the starting point for a processor register.

Supplier	Role	Holds personal data?
Supabase	Database, auth, storage, functions	Yes — all of it
Vercel	Web hosting, serverless functions	In transit only
Paynow	Payment processing	Yes — payment details
DigitalOcean	Paynow relay	In transit only; stores nothing
Expo (EAS)	Builds, updates, push delivery	Push tokens
Namecheap	Domains, DNS, mailbox	Contact form emails
Google	Play distribution, optional sign-in	Name and email if sign-in used
GitHub	Source code, legal pages	No

Prepared from the live system — database schema, deployed functions, scheduled jobs and application source — rather than from recollection.